

PROJECT DESIGN PHASE

Solution Architecture

Solution Architecture

The proposed architecture defines how employee information moves from an Excel spreadsheet into the ServiceNow employee table and how the resulting data is made available for review. The design separates data intake, staging, transformation, record matching, storage and reporting so that each part has a clear role.

Architecture and Data Flow

Figure 1: Solution architecture and data flow for the ServiceNow employee data import project.

Component Responsibilities

Architecture Flow

- 1. Data Source: Employee information is prepared in Excel.
- 2. Data Intake: The spreadsheet is loaded into ServiceNow through Load Data.
- 3. Staging: The incoming rows are held in the Employee Import table.
- 4. Transformation: The Transform Map maps the source fields to the Employee Test table.
- 5. Record Matching: Coalesce checks Employee ID to identify an existing employee.
- 6. Target Storage: New employees are inserted and matching employees are updated.
- 7. Visibility: Reports and the dashboard present the resulting employee information.

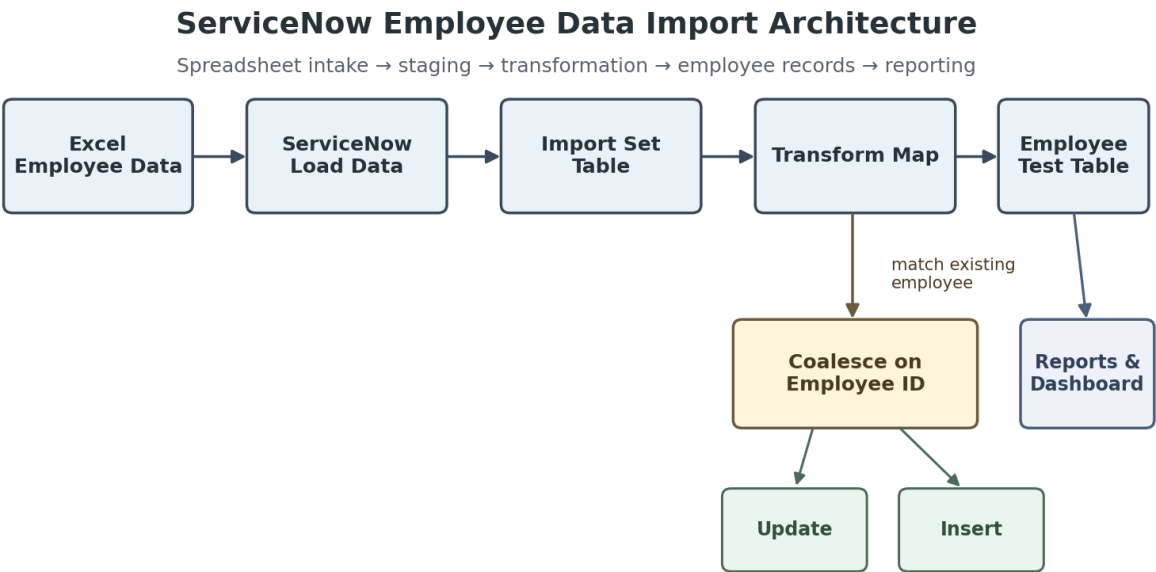
Architecture Characteristics

Architecture Scope Note

This architecture represents the completed academic project scope. It does not assume external enterprise integrations, scheduled production imports, load balancing or measured production-scale performance because those features are not part of the supplied project implementation.

Reference

Supplied Solution Architecture template, Project Design Phase.



Date	03 October 2026
Team ID	SWTID-2026-8740
Project Name	Employee Data Management using ServiceNow Import Sets and Transform Maps

Date	03 October 2026
Maximum Marks	4 Marks

Component	Role in the Solution	Project Implementation
Excel Employee Data	Provides the source employee information for an import.	Employee ID, Employee Name, Email, Department and Location.
ServiceNow Load Data	Loads the spreadsheet and starts the import process.	Used to load the Excel file into ServiceNow.
Import Set Table	Acts as the staging area for incoming records before transformation.	Employee Import table.
Transform Map	Controls how source fields are mapped into the target table.	Sample Spreadsheet Import transform map.
Coalesce	Matches incoming data with an existing employee so the record can be updated instead of duplicated.	Employee ID used as the matching value.
Employee Test Table	Stores the transformed employee records.	Target table: Employee Test.
Reports	Summarize employee information for review.	Employees by Department, Employees by Location and Employee List Report.
Dashboard	Combines the reports into one monitoring view.	Employee Analytics Dashboards.

Characteristic	Design Consideration
Modularity	The workflow separates staging, transformation, record matching, storage and reporting.
Data Integrity	Coalesce provides a defined matching mechanism for existing employees and reduces duplicate creation during repeated imports.
Maintainability	Field mappings are managed through the Transform Map rather than requiring manual entry for every employee.
Usability	Reports and the dashboard provide a simple way for administrators to review employee information.
Extensibility	The design can be extended with additional fields, validation rules, scheduled imports and access controls as requirements grow.